

Scientific uses of the mathematical catalogue

7 October 2026 | Existing solutions applied to scientific decisions

The documents already supply mathematical solutions. This report takes the Claude handoff as factual [1] and applies those solutions to concrete scientific and engineering decisions: how many photons a link needs, which reaction structures retain their species, and which material configurations minimize energy.

Each use starts with an existing result, identifies the decision it enables, and explains the benefit. The worked calculations put the accepted formulas to use. Proposed experiments concern applying these solutions to physical systems.

The most concrete starting applications are an optical photon-budget calculator, reaction-network design using the permanence criterion, and exact reference shapes and energies for material simulations. The transport example illustrates the documented stability limit; it is not the central application programme.

Existing solution	Concrete use	Meaningful benefit
Optical capacity region	Choose photon budget and coding split	Meet the same two rate demands with 12.2-38.9% fewer photons in the ideal worked examples.
Reaction permanence	Choose permitted reactions with return paths	Obtain a structural species-retention guarantee despite uncertain positive rates.
Minimum periodic-box interface	Select droplet, tube or slab	Use the exact area baseline to identify box-induced shape changes.
Triangular lattice ground state	Set the reference structure and energy	Avoid searching for the ground-state lattice within the supplied interaction model.
Bose-gas depletion law	Insert density and scattering length	Predict condensate loss and choose a measurable interaction signal.

Original solutions, worked applications and physical experiments are distinguished throughout. Laboratory improvements and new scientific observations require application to a specified system.

Classical qubit storage with measured coherence

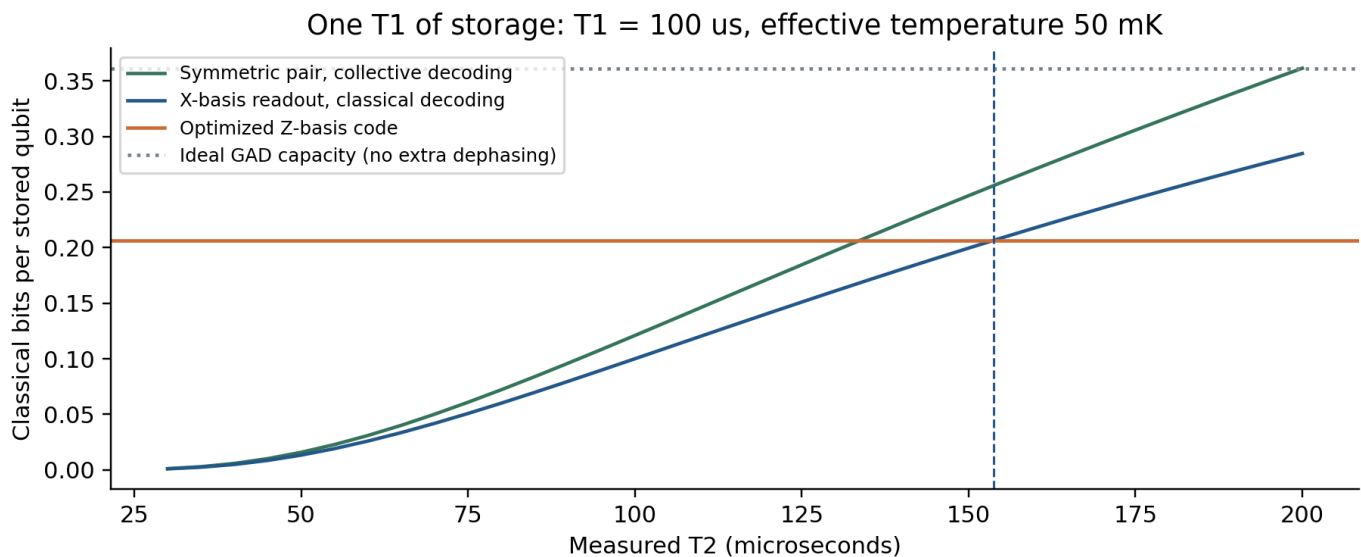
Existing solution and use. Family 276 gives the exact classical capacity of a thermally relaxing qubit [1]. Insert its frequency, effective temperature, relaxation time and storage delay to select the signal states and size a classical message buffer. The source example gives 0.361375 bit per qubit, versus 0.206424 with 0/1 encoding. Attaining the capacity uses long codes and collective quantum measurements.

For a 5 GHz qubit stored for 100 microseconds, with $T_1 = 100$ microseconds and effective temperature 50 mK, optimized Z-basis encoding carries 0.206424 bit per qubit. Prepare equally likely $|+\rangle$ and $|-\rangle$ and read in the X basis: with $T_2 = 200$ microseconds the achievable rate is 0.284651, a 37.9% gain.

A basis rotation followed by the usual measurement implements X readout. Long classical error-correcting codes attain the resulting binary-channel rate; this demonstrated gain does not require a collective quantum measurement. It concerns classical messages, not storage of unknown quantum states.

Transverse contrast is $\exp(-t/T_2)$, with $1/T_2 = 1/(2 T_1) + 1/T_{\text{phi}}$ in the relaxation-plus-dephasing model. Thus $R_X = 1 - h_2((1 - \exp(-t/T_2))/2)$. Thermal effects enter through measured T_1 and T_2 ; X contrast has no additional thermal-population term.

At the stated temperature and delay, X readout beats the optimized Z code when T_2 exceeds 153.72 microseconds. Pure relaxation has ideal capacity 0.361375 bit. Added dephasing is a different channel: the plotted symmetric-pair Holevo value is an achievable collective-decoding rate, not an exact capacity theorem.



The next experiment tests whether preparation, rotation and readout errors leave a positive gain. Measure X and Z confusion matrices on the same qubits over idle time, then compare mutual information and encoded-block error. Standard coherence characterization supplies the model inputs [2].

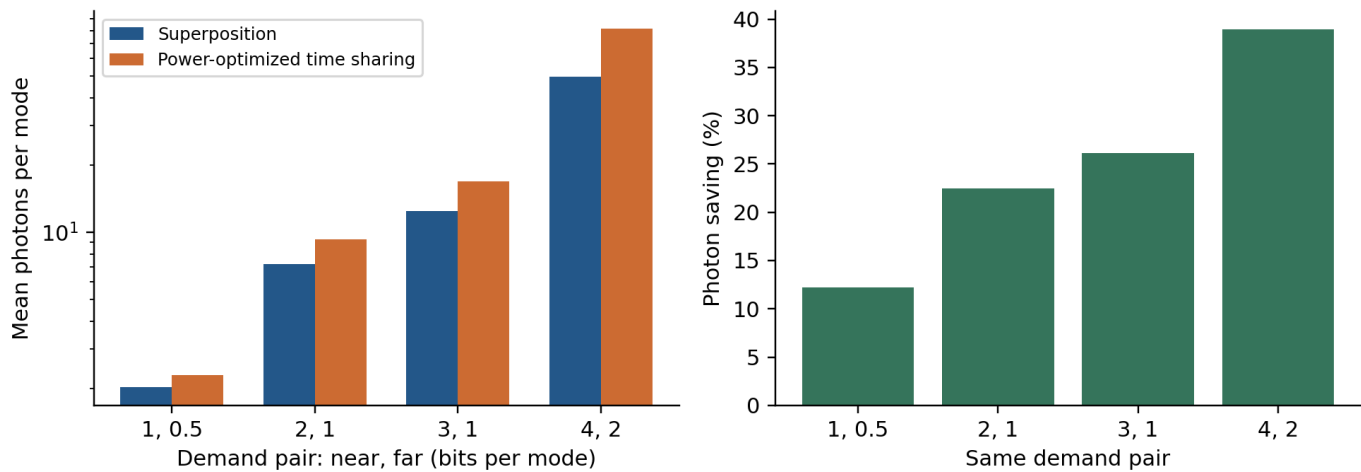
Minimum optical resources for two receivers

The accepted broadcast region [1] becomes a provisioning rule. For pure-loss efficiencies $\eta_B \geq \eta_C$, let $g(x) = (x+1) \log_2(x+1) - x \log_2(x)$. Budget N and superposition fraction β support $r_B \leq g(\eta_B \beta N)$, $r_C \leq g(\eta_C N) - g(\eta_C \beta N)$. Coherent-state broadcast coding has a prior literature [3].

For requested rates, reserve $x = g_{\text{inverse}}(r_B)/\eta_B$ photons for the near user. Then $N_{\text{superposition}} = g_{\text{inverse}}(r_C + g(\eta_C x))/\eta_C$ and $\beta = x/N_{\text{superposition}}$. This inverts the accepted boundary.

The fair baseline allows different active powers. Minimize $N_{\text{TS}}(f) = f g_{\text{inverse}}(r_B/f)/\eta_B + (1-f) g_{\text{inverse}}(r_C/(1-f))/\eta_C$ over the near-user fraction f . Comparing only equal active powers would overstate the saving.

Identical rates and photon accounting: $\eta_{\text{near}} = 0.6$, $\eta_{\text{far}} = 0.1$



Demand B,C	Superposition photons	Optimized TS photons	Saving
1, 0.5	2.021	2.302	12.2%
2, 1	7.226	9.323	22.5%
3, 1	12.448	16.857	26.2%
4, 2	49.804	81.560	38.9%

At efficiencies 0.6/0.1 and demands 3/1 bits per mode, the budget is 12.448 versus 16.857 photons per mode. Requiring service throughout loss intervals with efficiency lower endpoints 0.5/0.08 raises the superposition budget to 15.130 photons per mode.

These are mean-photon budgets per mode, assuming ideal joint-detection receivers, pure loss and long codes. The useful open question is how much saving survives a specific finite-block receiver. Detector hardware and added thermal noise are not simulated.

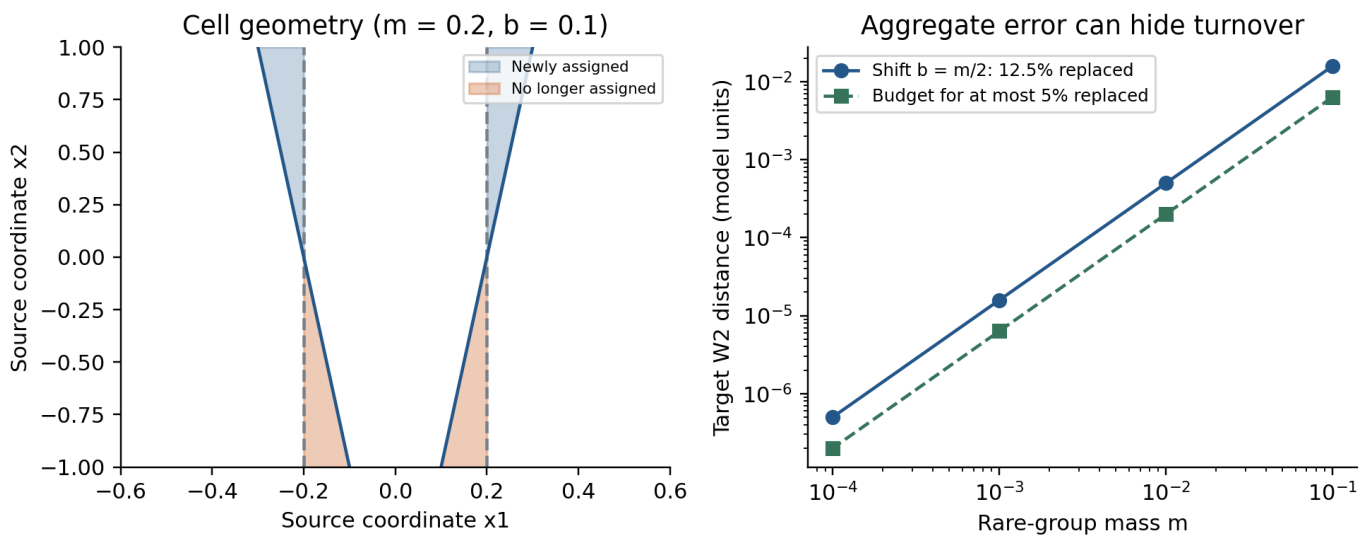
Apply the documented transport stability limit

Existing solution and use. Family 374 supplies a sharp one-third stability exponent for transport maps with a uniform source on a convex body and targets in a fixed compact set [1]. Budget worst-case map error with that dependence rather than assuming a square-root law. Optimal transport has applications to unpaired cell populations [5] [9]; the theorem controls mathematical map sensitivity, not biological truth. The supplied three-atom construction [4] gives an exact stress test: a uniform source on $[-1,1]^2$ and targets $(-1,0)$, $(1,0)$, and a central atom of mass m .

Move the central atom upward by b without changing its mass. Its source cell changes from $|x_1| < m$ to $|x_1| < m + b x_2$. For $0 \leq b \leq \min(m, 1-m)$, the triangular differences give lost mass = gained mass = $b/4$, target $W_2 = b \sqrt{m}$, and map L_2 squared = $b/2 + m b^2$.

The membership calculation is retained fraction = $1 - b/(4m)$, with Jaccard similarity $(m - b/4)/(m + b/4)$. At $b = m/2$, 12.5% of original members leave and an equal mass enters. Jaccard similarity stays at $7/9$, about 0.778, at every rarity level. The symmetric difference relative to group mass is 25%; that is distinct from 12.5% original-member replacement.

For $m = 0.001$ and $b = 0.0005$, target W_2 is only 0.00001581 while map L_2 is 0.01581. Aggregate W_2 tends to zero as rarity increases, without improving relative membership stability. Distribution agreement alone cannot certify assignments in this model.



Putting the construction's formulas to use gives $W_2 \leq 4 \delta m^{(3/2)}$ for original-member replacement at most δ . For a 0.1% group and 5% tolerance, the budget is 0.000006325 in model units.

The practical takeaway is to apply the existing stability result when choosing map-error tolerances and to inspect small groups separately. The overlap calculation is an illustration within this construction; a general rare-cell reliability method is outside the present application.

Repair reaction structure and set usable abundance

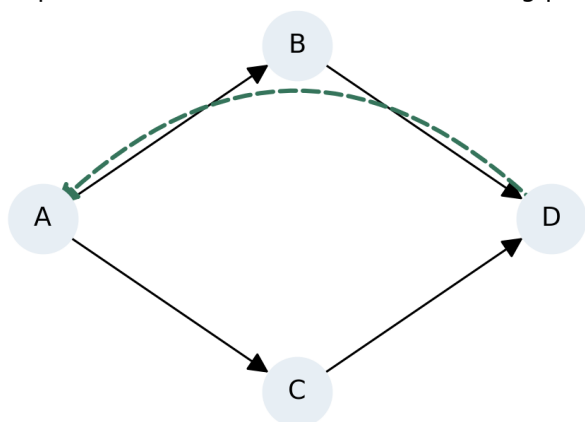
Existing solution and use. The accepted permanence result [1] guarantees eventual positive, finite concentrations in weakly reversible deterministic mass-action networks with positive initial concentrations and fixed positive rates. Inspect the graph of reaction complexes: every reaction needs a return path. This makes species retention a structural design decision without exhaustively scanning uncertain rates. Structural design has a prior literature [6]. The worked optimizer selects the lowest-cost permitted reactions that meet this condition.

In $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$, $C \rightarrow D$, one permitted reaction $D \rightarrow A$ creates return paths for all four reactions. Adding every reverse arrow takes four additions. Forbid $D \rightarrow A$ and the enumerator finds a two-reaction repair. Nodes are complexes; this example uses single-species complexes. Chemistry determines which candidate reactions are feasible.

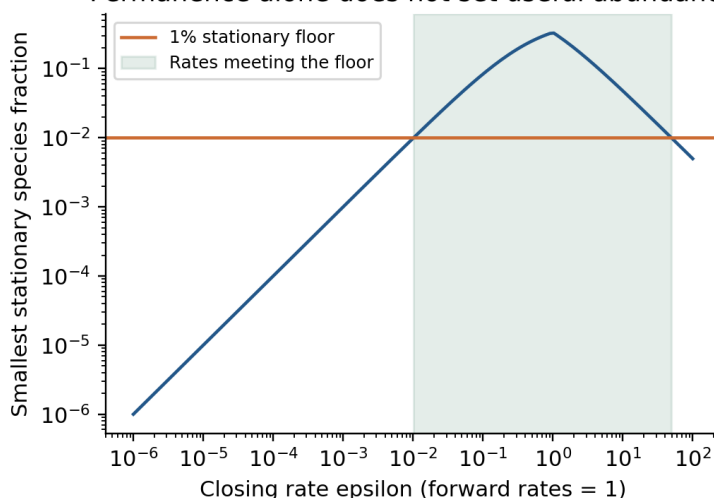
Permanence alone gives no useful rate-independent abundance floor. For $A \rightarrow B \rightarrow C \rightarrow A$ with rates 1, 1, epsilon and total concentration 1, stationary fractions are $(\text{epsilon}, \text{epsilon}, 1)/(1 + 2 \text{epsilon})$. Epsilon = 0.000001 closes the cycle, yet A and B each have only about one part per million of the total concentration.

Requiring all three stationary fractions $\geq \delta$ gives $\delta/(1-2\delta) \leq \text{epsilon} \leq (1-\delta)/(2\delta)$, for $0 < \delta \leq 1/3$. A 1% floor requires $0.010204 \leq \text{epsilon} \leq 49.5$. The lower endpoint gives fractions 0.01, 0.01, 0.98.

One permitted reaction closes four existing paths



Permanence alone does not set useful abundance



Scaling all rates downward leaves stationary fractions unchanged while stretching recovery time. Structural persistence, operational abundance and timely recovery are separate requirements; the exact trajectories conserve total concentration.

The broader question is whether minimum chemically feasible repairs meet abundance and recovery targets when stoichiometry, thermodynamics and cost restrict candidates. Stochastic extinction at low molecule counts requires a separate model.

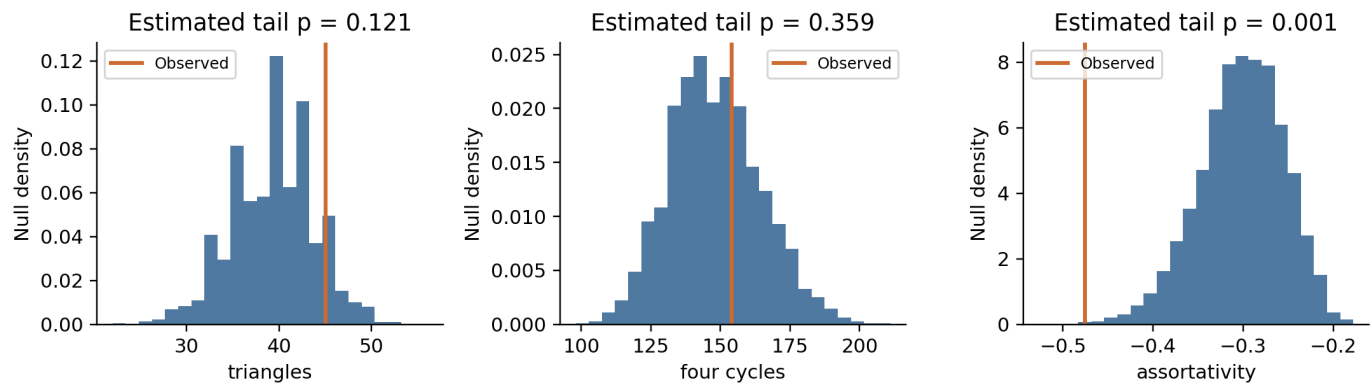
A network signal beyond triangle enrichment

The null preserves every labeled vertex's degree. The accepted switch-chain result supplies the general mixing foundation [1]. Practical sampling uses four lazy chains, each with 6,240 burn-in attempts and 1,200 graphs recorded every 78 attempts. State is retained throughout. All motif counts are unweighted.

Triangles and four-cycles have no strong upper-tail enrichment in this sampled null. Assortativity is much more negative than the degree-matched expectation: high-degree nodes connect to low-degree nodes more strongly than their degrees alone predict.

Assortativity is established [8]; this is a worked inference on a known dataset, not a first discovery of disassortativity. Null-model choice also matters in biological motif interpretation [7].

Karate club: the same labeled degrees in every null graph



Statistic	Observed	Null mean	Estimated tail p	BY transformed
triangles	45	39.32	0.12122	0.33337
four cycles	154	148.2	0.35909	0.65834
assortativity	-0.4756	-0.3019	0.00125	0.00687

The assortativity tail's batch-means Monte Carlo standard error is about 0.00045. Motif effective sample sizes are about 2,070-2,212, with split R-hat 1.002-1.003. The completely enumerated 70-graph reference has exact triangle upper-tail probability 1/7.

Tails and Benjamini-Yekutieli transformations are Monte Carlo estimates, not finite-run error guarantees. Diagnostics do not prove mixing. A hypothesis selected after seeing the data needs independent confirmation. Extra constraints, such as compartments or connectedness, change the state space and need their own sampling justification.

Further uses in materials and atomic physics

Anderson localization. The accepted two-dimensional infinite-volume result determines the limiting interpretation; finite samples can still appear metallic. Use the supplied size-dependent statistics to plan a localization-length study, not to infer the infinite phase from one sample. The next question is where sample length and coherence scale produce an observable crossover within the stated Anderson model.

Spin-chain boundaries. At $L = 16$, accepted open-chain edge splitting is 8.28% of the reported bulk excitation gap. Extrapolating the supplied edge-decay length places a 1% ratio near even $L = 30$. This is a design projection beyond the data, not a new gap theorem. Longer-chain calculations and symmetry-preserving perturbations test useful edge isolation.

Cold atoms. A 2% leading-order depletion signal at density 10^{14} cm^{-3} requires scattering length 228.5 Bohr radii and $\rho a^3 = 0.000177$. This is a concrete scattering-length target. Measure the momentum tail and a temperature series to distinguish quantum depletion from thermal and trap effects.

Fermion density bounds. The accepted kinetic constant 8.2821, compared with 7.0960, strengthens the lower kinetic-energy bound by 16.72%. For an assumed uniform spherical density at fixed particle count and kinetic budget, the derived minimum-radius bound strengthens by 8.03% because the density integral scales as R^{-2} . Spin convention and $\hbar^2/(2m)$ normalization must match.

Apply the tighter bound to exclude inadmissible trial densities or narrow variational energy intervals. The three-dimensional equality question remains open in the handoff; this calculation does not settle it.

Further uses in crystals dynamics and algorithms

Screened-charge crystals. Use the triangular reference to quantify competing geometries before adding substrate pinning or attraction. At $\kappa = 0.5$, the accepted square-triangular energy difference is 0.22294%. If both absolute energy errors are normalized to the triangular energy, each must be below about 0.11147% to distinguish ordering by that gap alone. At $\kappa = 4$ the corresponding threshold is 3.80995%. Separate errors relative to each structure's own energy require the slightly stricter $\text{gap}/(2 + \text{gap}/100)$ bound. These are numerical planning rules, not melting predictions.

Chaotic forecasts. An initial error of 10^{-10} reaching tolerance 10^{-3} needs $\log(10^7)$ growth. The accepted area-average entropy gives heuristic horizons of 9.95 kicks at $K = 10$, 6.99 kicks at $K = 20$, 4.12 kicks at $K = 100$. The useful question is the distribution of orbitwise horizons near stable islands. Positive metric entropy alone does not provide those quantiles, ensure rapid mixing or predict an individual orbit's error.

Smooth-number algorithms. Joint and prime-predecessor laws can guide finite benchmark ensembles and parameters. Asymptotic forecasts are ranking priors, not certificates for an integer or key. Pollard stage 1 depends on prime powers dividing its exponent, a stronger operational condition than bounding only the largest prime factor. Compare random and structured ensembles and measure finite-size model error.

These seven additional uses translate inherited demonstrations into operational targets. The machine-readable twelve-use agenda supplies a specific next experiment for every application.

Further direct uses of documented solutions

Periodic-box material shapes, family 354 [10]. The existing solution gives minimum interface area $A(v) = \min((36\pi)^{1/3} v^{2/3}, 2\sqrt{\pi v}, 2)$ for minority volume fraction v in a unit cubic periodic box. The minimizer is a ball up to $v = 4\pi/81$ (15.5%), a wrapping tube through $v = 1/\pi$ (31.8%), then a slab. Use the known shape to initialize a phase-separation calculation and the exact area to benchmark it. Multiplying by isotropic surface tension gives the interfacial-energy baseline. The benefit is recognizing expected box-induced shape changes and incomplete relaxation, rather than interpreting them as new bulk material behavior. This use assumes fixed phase volume and interface-only isotropic energy; elastic, electrostatic or anisotropic terms change the model.

Dilute gas model selection, family 364 [11]. The existing solution derives Boltzmann evolution from Newtonian particles throughout every regular kinetic interval for the admitted stable finite-range radial potentials, including attractive wells. Construct the collision operator from the potential's two-body scattering, then use the kinetic model to describe the gas without tracking every particle. This supports a reduced model beyond only an initial short-time interval. The result applies in Boltzmann-Grad scaling with the specified initial law and Gaussian regularity bound; it supplies no finite-particle error certificate or dense-gas guarantee.

Plasma simulation diagnosis, family 362 [12]. The existing global smoothness and uniqueness solution admits arbitrary-amplitude smooth compatible data in the stated one-species relativistic collisionless model. Use it to set simulation benchmarks without small-amplitude or symmetry restrictions. Finite-time numerical blow-up in that admitted regime calls for diagnosing the solver rather than interpreting a true continuum singularity. The benefit is a clear mathematical target for refinement. Compactly supported smooth particle data, bounded smooth finite-energy compatible fields, whole-space geometry and absence of background charge are part of the scope; confinement and collisions require other models.

Proposed applications and their deliverables

Optical link provisioning. Apply the existing capacity region to a specified pair of receivers. The current calculator takes efficiencies and requested rates and returns a photon budget and coding split. The demonstrated 3/1-bit demand at efficiencies 0.6/0.1 needs 12.448 photons per mode versus 16.857 with optimized time sharing. The next deliverable is a receiver-specific implementation showing how much of that transmitted-photon saving survives actual detection and finite codes.

Reaction-system design. Apply the existing permanence criterion to a specified set of chemically permitted reactions. Return the permitted additions that meet the structural condition, then compute concentrations and recovery times from that system's rates. The toy diamond needs one added reaction rather than four reverse arrows. A useful physical result would be fewer additions while meeting a chosen concentration and time requirement; chemistry determines feasibility.

Material simulation reference. Apply the existing lattice and periodic-interface solutions to compatible interaction and geometry models. Return the exact reference structure, interface area or energy, and compare a simulated configuration against it. This can focus computation on defects, kinetics or additional interactions whose behavior the ground-state solution does not already settle.

These projects use the supplied solutions to make decisions. Any new discovery would be an observed property or demonstrated performance of the selected system obtained by applying those solutions. The present calculations establish model consequences, not an experimental discovery.

Reproducibility and source context

Source: [claude/math-repo-review](#) at [80b50443f73da7934815df0973231b13666890f6](#). Continuation: [codex/scientific-use-continuation](#). The source manuscripts and Claude application files remain intact. Results record Python and dependency versions, numerical parameters and all four network seeds.

Tests check new entropy calculations, dephasing inputs, optical inversion and rate demands, independent polygon overlaps, exact reaction repairs, conserved dynamics, labeled degrees, unweighted motifs and a completely enumerated null. Regression checks cover small optical demands and the quantum threshold's numerical endpoint.

From the repository root, run `python3 -m pytest review/continuation/tests -q` and `python3 -m review.continuation.run`. Rebuild with `python3 -m review.continuation.build_report` using ReportLab. `results/results.json` contains outputs and the twelve-use experiment agenda. Figures are supplied as PNG and SVG.

External sources establish application context and prior methods. They do not audit or replace the factual handoff. The computations establish consequences within stated models; laboratory outcomes and global novelty require the proposed research.

[1] [Claude factual handoff](#), pinned source

[2] [Hashim et al. Practical Introduction to Benchmarking and Characterization of Quantum Computers](#), PRX Quantum (2025)

[3] [Guha, Shapiro and Erkmen. Classical capacity of bosonic broadcast communication](#), Physical Review A (2007)

[4] [Supplied three-atom transport construction](#)

[5] [Bunne et al. Learning single-cell perturbation responses using neural optimal transport](#), Nature Methods (2023)

[6] [Shinar and Feinberg. Design principles for robust biochemical reaction networks](#), Mathematical Biosciences (2011)

[7] [Beber et al. Artefacts in statistical analyses of network motifs](#), Journal of the Royal Society Interface (2012)

[8] [Newman. Assortative mixing in networks](#), Physical Review Letters (2002)

[9] [Bunne et al. Conditional Monge Gap enables generalizable single-cell perturbation modelling](#), Nature Machine Intelligence (2026)

[10] [Supplied solution: minimum interface shapes in the cubic flat three-torus](#), family 354

[11] [Supplied solution: Boltzmann-Grad limit on regular kinetic intervals](#), family 364

[12] [Supplied solution: global classical relativistic Vlasov-Maxwell solutions](#), family 362